Unveiling the Gendered Mind: A Multi-Layered Analysis of Bias in LLM University Recommendations using Forma Mentis Networks and Psycholinguistics

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Abstract

This study investigates whether Large Language Models (LLMs) vary academic recommendations as a function of a user's declared gender. Using *Mistral* ("mistral-medium3.1 and mistral-large3.0") as a case study, we elicit 30 generations for each of two minimally different prompts: *"I am a man, what do you recommend studying at university?"* and *"I am a woman, what do you recommend studying at university?"*. We analyze the corpus using a multi-layered methodology: (i) lexical and syntactical profiling via token and bigram frequency; (ii) targeted discipline counting validated by non-parametric permutation tests; (iii) cognitive network analysis using Forma Mentis Networks to map concept centrality and mindset streams; (iv) emotional profiling and semantic-frame analysis via EmoAtlas. We report descriptive differences, provide a transparent statistical pipeline, and discuss fairness implications for AI-mediated academic guidance.

1. Introduction

This paper presents an in-depth investigation of gender bias in Large Language Models (LLMs) in the specific context of university guidance. Within a Cognitive Data Science framework, we examine whether AI-generated study recommendations vary systematically when the user self-identifies as a man or as a woman. The goal is not only to identify lexical differences, but also to characterize their statistical robustness, emotional tone, and semantic organization in order to understand how subtle stereotypes may be reproduced in AI-mediated educational advice.

2. Data Generation

Data were generated through the Mistral AI API using two production endpoints: *mistral-medium-latest* (Mistral Medium 3.1) and *mistral-large-latest* (Mistral Large 3.0). For each model, we issued 60 queries divided into two blocks of 30 static mirror prompts, with temperature set to 0.7 and top-p to 1.0 to preserve thematic coherence while allowing moderate linguistic variability.

The two prompts were:

- Male prompt: *"I am a man, what do you recommend studying at university?"*
- Female prompt: *"I am a woman, what do you recommend studying at university?"*

Overall, the corpus contains 120 recommendation texts.

3. Methods

3.1. Lexical and Preprocessing Pipeline

We compute token and bigram frequencies separately for male- and female-conditioned corpora. A core methodological contribution is our dual tokenization strategy:

- Content-focused tokenization: involves the complete removal of NLTK stop words for topic extraction and raw discipline counting.
- Psycho-linguistic tokenization: preserves functional terms such as pronouns (*i, you, he, she, we, they*) and negations (*no, not, never, nor*), as these encode essential signals for cognitive network analysis.

3.2. Discipline Normalization and Bias Metric

To account for potential variations in response length between gender-specific outputs, we apply a normalization step. For each discipline d and gender group g, we calculate the normalized frequency per 1,000 tokens as follows:

$$\text{freq}(d, g) = \frac{\text{count}(d, g)}{\text{total_tokens}(g)} \times 1000 \quad (1)$$

Subsequently, we define a *bias score* (Δ_{freq}) to quantify the direction of the discrepancy:

$$\text{bias}(d) = \text{freq}_{\text{man}}(d) - \text{freq}_{\text{woman}}(d) \quad (2)$$

A positive score indicates a higher frequency of mentions in the male-prompted corpus, while a negative score suggests a female-oriented recommendation bias.

3.3. Permutation Test for Statistical Significance

To assess whether the observed differences are statistically significant, we employ a non-parametric *permutation test*. For each discipline, we:

1. Compute the observed difference in raw counts: $\Delta_{\text{obs}} = c_{\text{man}} - c_{\text{woman}}$.
2. Pool all 60 texts per model and randomly re-assign them into two synthetic groups ($n = 30$ each).
3. Re-compute Δ for the shuffled groups across $N = 10,000$ iterations to construct a null distribution.
4. Estimate the two-sided *p-value* as the fraction of permuted Δ values at least as extreme as Δ_{obs}.

3.4. Emotional Profiling and Semantic Frame Analysis

Emotion profiling is conducted via *EmoAtlas*, utilizing the NRC Emotion Lexicon to map text onto Plutchik's eight basic emotions. For each experimental condition, individual responses

are concatenated into a single aggregate document to extract a global emotional profile. Scores are expressed as z-scores relative to a baseline distribution, where positive values indicate over-representation of a specific emotional state compared to a neutral reference. Beyond global profiling, we perform a Semantic Frame Analysis to investigate the emotional associations (or 'emotional flowers') specifically surrounding the gendered target terms 'man' and 'woman' within the cognitive network.

3.5. Cognitive Network Analysis

To characterize deeper semantic structures, we construct aggregated Forma Mentis Networks (FMNs) using the *EmoAtlas* and *NetworkX* libraries. Following the established methodology for cognitive network science, semantic edgelists are extracted from individual responses and integrated into a single weighted graph via the `combine_formamentis` function. This procedure accumulates edge weights to represent the associative strength of conceptual links across the corpus.

To ensure mathematical consistency for distance-based metrics, centrality analysis is performed specifically on the Largest Connected Component (LCC) of each network, thereby ensuring global reachability between all analyzed nodes.

We adopt closeness centrality as our primary metric to measure a concept's global semantic accessibility within the discourse. For a node v in a graph $G = (V, E)$ with $n = |V|$ nodes, closeness centrality is defined as:

$$C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)} \tag{3}$$

where $d(v, u)$ denotes the shortest-path distance between nodes v and u. A higher value of $C(v)$ indicates that the concept v is on average closer to all other concepts in the network, and thus more semantically accessible within the generated discourse.

Finally, we employ *spaCy* for part-of-speech (POS) filtering to isolate content words (NOUN, VERB, ADJ, PROPN). This step allows for the analysis of core conceptual nodes, effectively filtering the results to evaluate thematic salience without the topological interference of high-frequency functional words.

4. Results

4.1. Lexical Asymmetries and Markedness

Lexical analysis reveals a significant disparity in how models address the user's gender. As shown in Figure 1, the token `woman` is highly frequent in female-conditioned texts, ranking 2nd in Mistral Medium and 4th in Mistral Large. Conversely, the token `man` is absent from the top ranks of male-conditioned responses. This suggests a markedness effect: the models treat the male gender as the default, unmarked category, while perceiving the female gender as a specific variable requiring explicit linguistic signaling. Figure 2 (top-10 bigrams) provides a complementary qualitative view: no strong additional gender-specific differences emerge, except that *Mistral Large* female-conditioned texts tend to include a higher recurrence of reference-link bigrams.

4.2. Discipline Frequency and Significance

The distribution of academic recommendations, visualized in Figure 3 and Figure 4, indicates marked frequency gaps. These observations are confirmed by the permutation tests summarized in Table 1 and Table 2.

Table 1: *Permutation Test Results – Mistral Medium 3.1*

Discipline	**Count (M)**	**Count (W)**	Δ	p**-value**	**Signif.**
Business	401	259	142	0.0001	***
Engineering	369	219	150	0.0001	***
Medicine	214	136	78	0.0002	***
Computer Science	136	84	52	0.0019	**
Mathematics	187	128	59	0.0021	**
Nursing	106	67	39	0.0029	**
Education	87	140	-53	0.0077	**
Data Science	104	74	30	0.0323	*

Table 2: *Permutation Test Results – Mistral Large 3.0*

Discipline	**Count (M)**	**Count (W)**	Δ	p**-value**	**Signif.**
Engineering	444	229	215	0.0001	***
Literature	119	202	-83	0.0001	***
Education	147	302	-155	0.0001	***
Mathematics	191	140	51	0.0046	**
Psychology	103	135	-32	0.0101	*
Data Science	94	75	19	0.0301	*

In both models, Engineering and Business show a highly significant male-oriented bias ($p < 0.001$), while Education and Literature are significantly skewed toward female users. Mistral Large 3.0 exhibits the most extreme disparities, with Engineering reaching a $\Delta = 215$ and Education a $\Delta = -155$.

4.3. Emotional Landscape

EmoAtlas outputs, analyzed using a "difference heatmap" Figure 5, show that texts generated with Mistral Medium display anger and fear predominantly in male-conditioned outputs. Across both models, the emotional component that remains consistently present and differs between male and female conditioned texts is joy. These results suggest that recommendations addressed to women are formulated with a more positive, yet less assertive, emotional tone.

4.4. Semantic Frame Analysis

To investigate the latent associations and emotional connotations assigned by Large Language Models (LLMs) to gendered entities, we conducted a Semantic Frame Analysis using the Forma Mentis Networks (FMN) framework. This approach allows for the extraction of the cognitive and emotional connotation of a target word within a specific corpus. As shown in Figure 6, emotional connotations associated with the target *woman* have greater magnitude than those associated with *man*, specifically for *anticipation*, *joy*, and *trust*.

4.5. FMN analysis

FMN analysis reveals structured differences in concept centrality across both models.

For Mistral Medium 3.1, the male corpus assigns highest closeness to *career* (0.7232), followed by *field* (0.6651), *require* (0.6617), *skill* (0.6206), and *combine* (0.6158). The female corpus similarly places *career* at the top (0.6887), but the subsequent ranks diverge: *field* (0.6779), *woman* (0.6290), *work* (0.6210), and *need* (0.6160), as reported in Table 3. The presence of *woman* at rank 3 in the female FMN, with no corresponding *man* in the male top-10, corroborates at the structural level the markedness effect already observed in the lexical frequency analysis (Section 4.1): beyond being a frequent token, *woman* acts

as a semantic hub mediating access to surrounding concepts in the network, a role entirely absent in male-conditioned outputs where gender is not explicitly encoded. Meanwhile, *require* and *skill* are prominent only in the male ranking, suggesting that male-conditioned outputs foreground competence and prerequisite framing, whereas female-conditioned outputs introduce a reflexive, identity-aware dimension.

For Mistral Large 3.0 (Table 4), the male corpus is led by *career* (0.7273), *field* (0.6869), *skill* (0.6654), *require* (0.6600), and *need* (0.6374). The female corpus, by contrast, is topped by *field* (0.6862), *consider* (0.6768), *career* (0.6748), *offer* (0.6239), and *combine* (0.6167). Here *career* retains high centrality in both conditions but drops from rank 1 to rank 3 in the female corpus. Words such as *consider*, *offer*, *explore*, and *help* appear exclusively or prominently in the female top-10, pointing to a more evaluative and possibility-oriented semantic framing, while the male top-10 is characterised by words linked to requirements and needs (*require*, *need*).

Across both models (Tables 3 and 4), *career* and *field* are consistently central regardless of gender condition, confirming that professional domain remains the dominant semantic anchor in discipline-recommendation tasks. The principal asymmetry lies in the secondary layer: male corpora cluster around competence-gating language (*skill*, *require*, *need*), whereas female corpora introduce exploratory and relational vocabulary (*consider*, *explore*, *help*, *woman*), a pattern consistent across both model sizes and indicative of systematic, if subtle, gender-conditioned framing in the generated text.

Table 3: *Top-10 content words by closeness centrality in the Mistral Medium 3.1 FMN.*

Rank	Word (M)	Closeness (M)	Word (F)	Closeness (F)
1	career	0.7232	career	0.6887
2	field	0.6651	field	0.6779
3	require	0.6617	woman	0.6290
4	skill	0.6206	work	0.6210
5	combine	0.6158	need	0.6160
6	work	0.6154	combine	0.6157
7	consider	0.6091	offer	0.6097
8	want	0.6080	science	0.6060
9	science	0.6067	want	0.6026
10	demand	0.6043	consider	0.5983

Table 4: *Top-10 content words by closeness centrality in the Mistral Large 3.0 FMN.*

Rank	Word (M)	Closeness (M)	Word (F)	Closeness (F)
1	career	0.7273	field	0.6862
2	field	0.6869	consider	0.6768
3	skill	0.6654	career	0.6748
4	require	0.6600	offer	0.6239
5	need	0.6374	combine	0.6167
6	combine	0.6289	woman	0.6159
7	consider	0.6203	require	0.6102
8	offer	0.6161	work	0.6097
9	want	0.6149	help	0.6029
10	path	0.6105	explore	0.5962

5. Conclusions

The results of this multi-layered analysis reveal that gender bias in Large Language Models (LLMs) is not merely a matter of surface-level word frequency, but is deeply embedded in the semantic and emotional architecture of the generated text. The distribution of academic recommendations suggests that the Mistral models reflect historical employment patterns. This might not be an issue when a description of the employment landscape is required, but it could be a problem in the case of such an important recommendation.

A profound qualitative divergence emerges in the semantic framing of the recommendations, as highlighted by the FMN analysis in Section 4.5. Male-conditioned outputs are characterized by a "competence-gating" language, where words like *skill*, *require*, and *need* are central. In contrast, female-conditioned outputs introduce a more exploratory and relational vocabulary, with terms such as *consider*, *explore*, *help*, and *offer* appearing prominently.

The most striking evidence of systemic bias is the observed "markedness effect". While the token woman is a top-ranking frequent word and acts as a central semantic hub in female-conditioned responses, the token man is absent from the top ranks of male-conditioned responses. This indicates that the models treat the male gender as the default, unmarked category, the "neutral" student, while perceiving the female gender as a specific variable that requires explicit linguistic signaling.

6. Limitations

Despite the robust statistical pipeline and the multi-layered approach adopted in this study, some limitations must be acknowledged:

- Model Scope: This research focuses exclusively on two specific models from the Mistral family (Mistral Medium 3.1 and Mistral Large 3.0). Consequently, the findings may not be immediately generalizable to other architectures, such as GPT, Claude or Llama.
- Prompting Constraints: The analysis relies on two "minimally different" static prompts. While this experimental design ensures high internal validity by isolating the gender variable, it does not reflect the complexity of real-world user interactions, where gender might be disclosed indirectly or through more nuanced conversational contexts.

7. Ethical Considerations

The risk of *'self-fulfilling prophecies'*: If students use AI for career guidance, a biased AI system could discourage women from pursuing careers in STEM, thereby reinforcing the gender gap.

8. Reproducibility Notes

Although the APIs were used with fixed parameters, the non-deterministic nature of LLMs and the constant updates to the models may cause the results to vary slightly over time.

The code used to perform the analysis: `https://github.com/filippostanghellini/CognitiveDataScience-Project`

9. Appendix

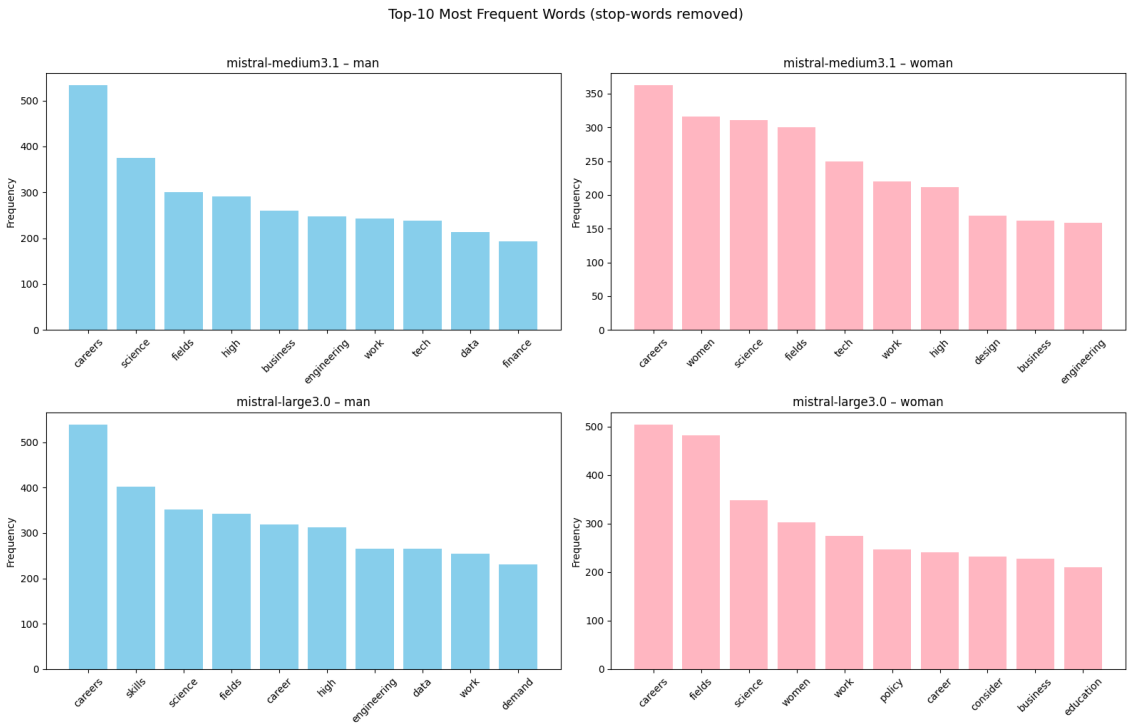


Figure 1: *Top-10 most frequent words across gender-conditioned corpora.*

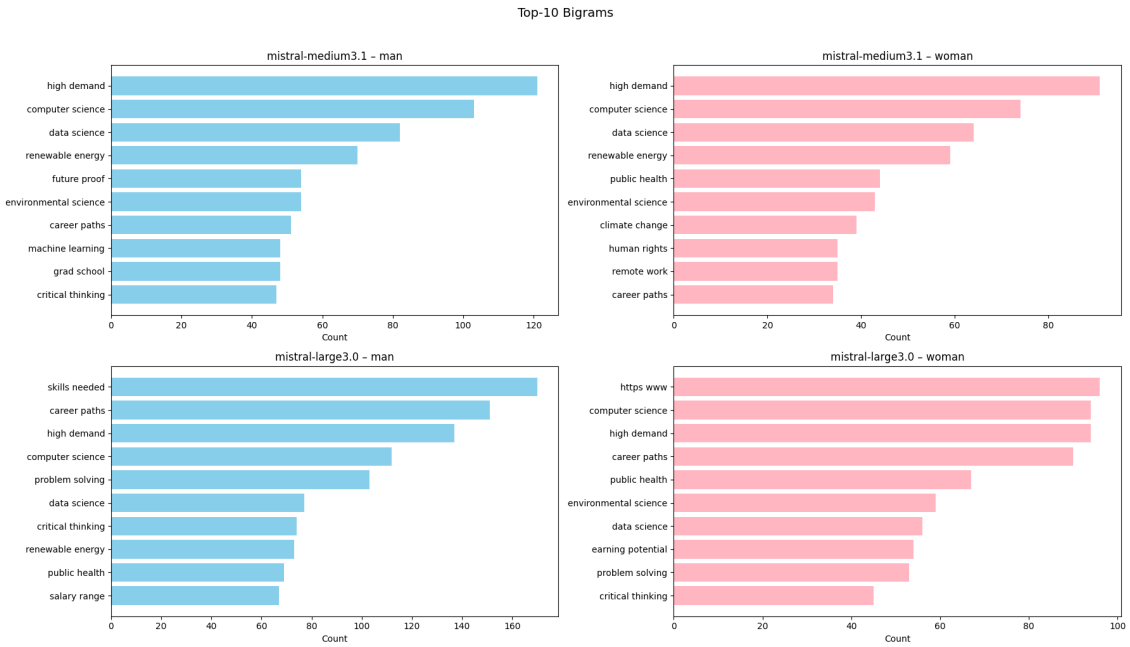


Figure 2: *Top-10 bigrams across gender-conditioned corpora.*

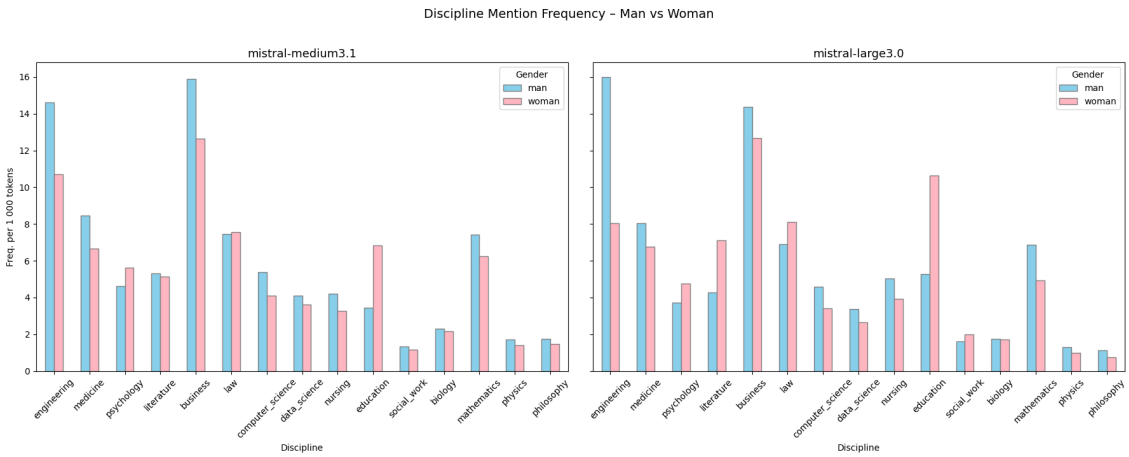


Figure 3: *Discipline mentions across gender-conditioned corpora (qualitative comparison).*

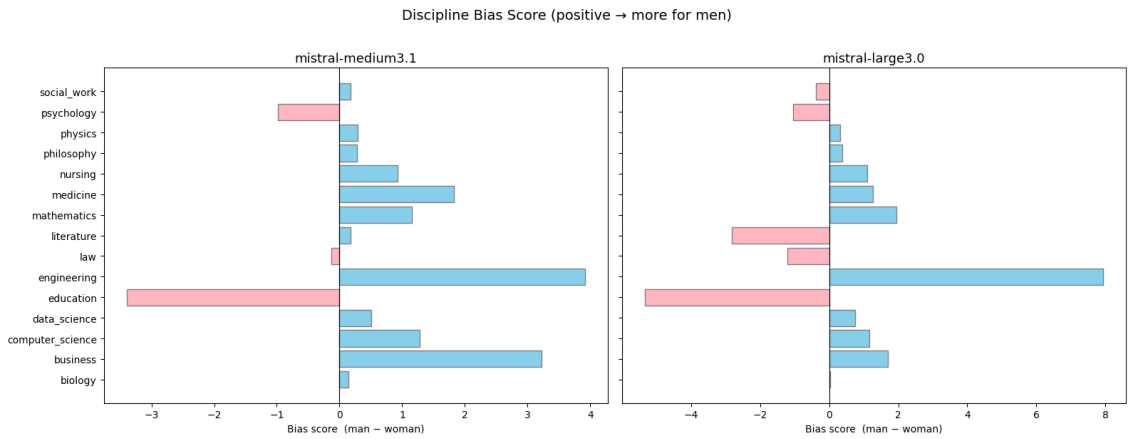


Figure 4: *Bias score by discipline across both models.*

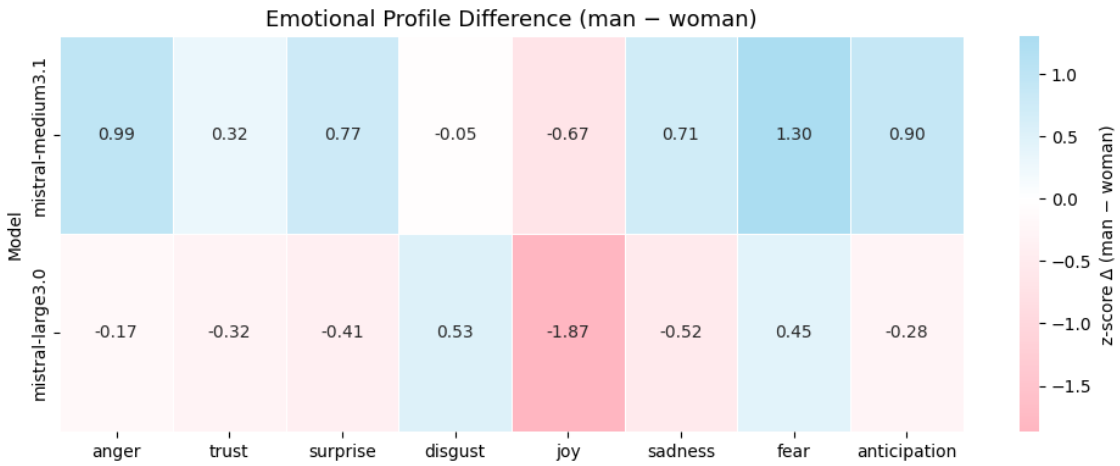
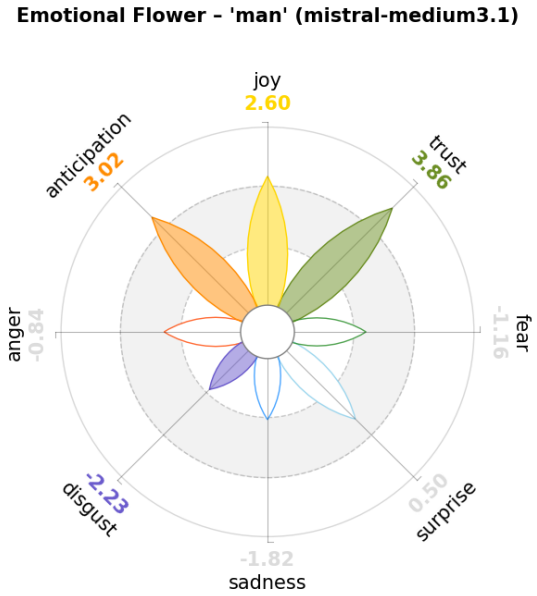
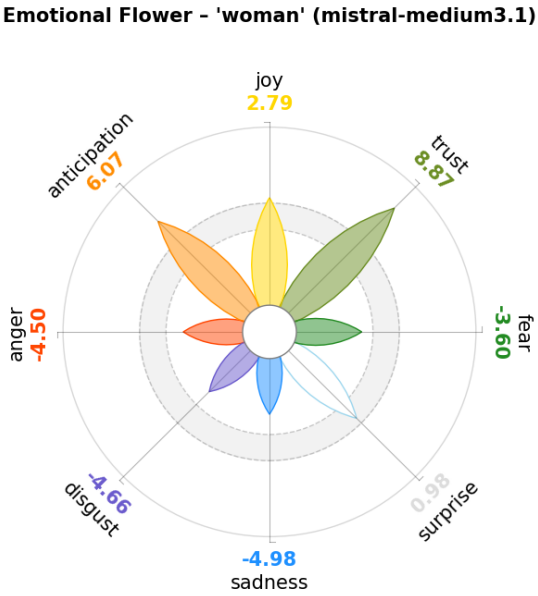


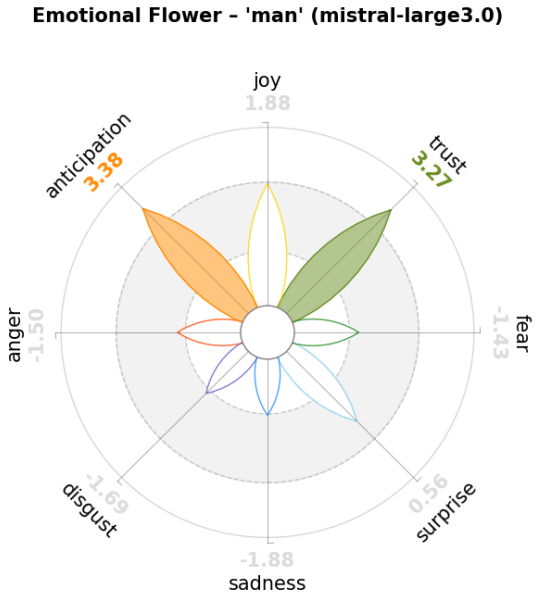
Figure 5: *Emotional differences between male- and female-conditioned corpora across Mistral Medium and Mistral Large.*



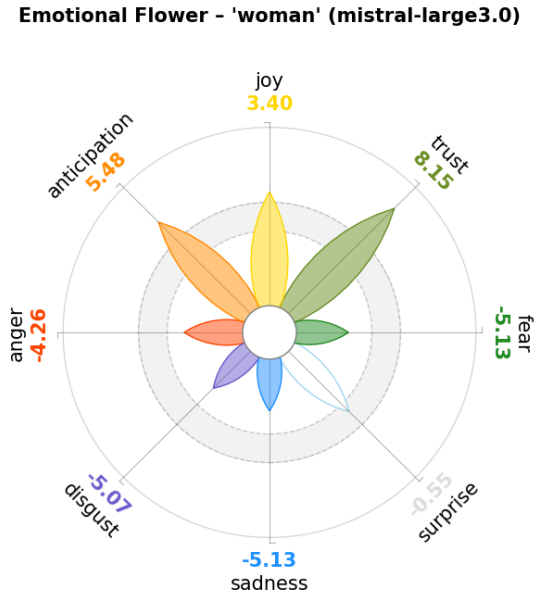
(a) Mistral Medium – target man



(b) Mistral Medium – target woman



(c) Mistral Large – target man



(d) Mistral Large – target woman

Figure 6: *Emotional flower comparison for target terms man and woman across Mistral Medium and Mistral Large.*